

Overview

A robot is a physical system that senses the environment, computes decisions, and acts through actuators. In applied engineering, robotics sits at the intersection of three disciplines:

1. mechanical and electrical systems,
2. control theory,
3. AI/perception/planning.

A compact architecture is:

- Sensors -> State Estimation -> Planner/Policy -> Controller -> Actuators -> Environment.

Relationship between AI, control, and robotics:

- AI handles perception, prediction, and decision-making in uncertain environments.
- Control handles stable execution of desired trajectories and setpoints.
- Robotics integrates both into real-time physical systems with safety constraints.

Kinematics & Dynamics (High Level)

Configuration Space, Joints, and Degrees of Freedom

Configuration space (C-space) represents all possible robot configurations. Each independent joint variable contributes one degree of freedom (DoF).

Examples:

- Differential-drive mobile base: typically 3 DoF in planar pose (x, y, heading).
- Industrial manipulator: commonly 6 DoF for full 3D pose control.

Forward vs Inverse Kinematics

- Forward kinematics (FK): joint values -> end-effector pose.
- Inverse kinematics (IK): desired end-effector pose -> joint values.

FK is usually straightforward; IK may have multiple, zero, or infinite solutions depending on geometry and constraints.

Dynamics Intuition

Dynamics connects motion and force:

- torque drives joint acceleration,
- inertia, friction, and gravity shape response,
- control inputs must account for these effects.

In real robots, the same command can behave differently as payload, friction, or battery state changes.

Control Basics

Feedback Control

Feedback compares desired state and measured state, then corrects error.

Error definition:

$$error(t) = target(t) - measured(t)$$

Why feedback matters:

- rejects disturbances,
- compensates for modeling inaccuracies,
- improves repeatability and robustness.

PID Controllers

PID combines three terms:

- P (proportional): reacts to current error.
- I (integral): corrects accumulated bias.
- D (derivative): damps rapid changes.

Control law:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

Stability and Response

Practical tuning focuses on:

- rise time,
- overshoot,
- settling time,
- steady-state error.

Too much P can overshoot, too much I can oscillate (integral windup), too little D can reduce damping.

Robot Architectures

Mobile Robots, Manipulators, Drones

- Mobile robots: logistics, delivery, inspection.
- Manipulators: assembly, welding, pick-and-place.
- Drones/UAVs: surveying, monitoring, delivery in constrained airspace.

On-board vs Off-board Compute (Edge vs Cloud)

On-board (edge) advantages:

- low latency,
- works offline,
- privacy/control.

Cloud/off-board advantages:

- larger model capacity,
- centralized coordination,
- easier fleet-wide updates.

Most production systems use hybrid architectures: safety-critical loops on edge, heavy analytics in cloud.

Business Case Studies & Exceptions

Case Study 1: Warehouse Mobile Robots

Problem:

- route goods efficiently while avoiding collisions and congestion.

System pattern:

- local feedback control for wheel velocity,
- global task allocation from fleet manager,
- edge inference for obstacle detection.

Trade-off:

- aggressive speed improves throughput but increases safety risk and navigation error.

Case Study 2: Industrial Manipulators in Manufacturing

Problem:

- high-precision repeated motion under variable payloads.

System pattern:

- calibrated kinematic model,
- tight feedback control,
- vision guidance for part alignment.

Exception:

- for ultra-high precision tasks, classical control with rigorous calibration may outperform complex AI controllers.

Interview Questions & Answers

1. Explain feedback control in simple terms.

It measures error between target and actual state, then continuously corrects actuator commands to reduce that error.

2. Forward vs inverse kinematics?

Forward maps joint values to pose; inverse computes joint values needed for a desired pose.

3. What is a degree of freedom?

An independent variable needed to define robot configuration.

4. What does the P term do in PID?

Applies correction proportional to current error.

5. What does the I term do?

Removes persistent steady-state error by accumulating past error.

6. What does the D term do?

Predicts error trend and adds damping to reduce overshoot.

7. What is overshoot?

When the response exceeds target before settling.

8. Why can integral windup be harmful?

Integral term accumulates excessively under saturation, causing instability/slow recovery.

9. **How does AI integrate into robotics?**
AI supports perception/planning, while control executes safe physical behavior.
10. **On-board vs cloud control trade-off?**
On-board gives low latency and reliability; cloud gives scale and heavier compute.
11. **Why is latency critical in robotics?**
Control loops must update quickly to remain stable and safe.
12. **What is C-space used for?**
Planning feasible motion while respecting robot geometry and joint limits.
13. **Why are models alone insufficient in robotics?**
Real hardware has disturbances, noise, and nonideal dynamics requiring feedback.
14. **What is a common robotics deployment failure?**
Good simulation performance but poor real-world robustness due to unmodeled dynamics.
15. **When should simple controllers be preferred over learned policies?**
In tightly bounded repetitive tasks where interpretability and certification matter.

References

- Modern Robotics specialization (mechanics/planning/control outcomes): <https://www.coursera.org/specializations/modern-robotics>
- MIT Underactuated Robotics notes: <https://underactuated.csail.mit.edu/>
- NVIDIA Edge AI and Robotics Teaching Kit syllabus: <https://developer.nvidia.com/edge-ai-robotics-teaching-kit-syllabus>
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